

Parameter-Efficient Fine-Tuning of BERT and Gemma using LoRA & QLoRA

Executive Summary

Large Language Models (LLMs) have achieved state-of-the-art performance across a wide range of Natural Language Processing (NLP) tasks. However, full fine-tuning of these models often requires substantial computational resources and GPU memory, making deployment difficult in resource-constrained environments.

This project investigates Parameter-Efficient Fine-Tuning (PEFT) techniques, specifically Low-Rank Adaptation (LoRA) and Quantized Low-Rank Adaptation (QLoRA), as practical alternatives to full fine-tuning.

Two experiments were conducted:

1. BERT Base Uncased fine-tuned using LoRA for sentiment classification.
2. Gemma 2B fine-tuned using QLoRA with 4-bit NF4 quantization.

The results demonstrate that strong classification performance can be achieved while updating only a very small fraction of model parameters. Furthermore, the Gemma 2B model was successfully adapted on a Tesla T4 GPU using QLoRA, highlighting the practicality of modern PEFT techniques.

1. Introduction

1.1 Background

Transformer-based models have become the foundation of modern NLP systems. Models such as BERT, GPT, LLaMA, Gemma, and Qwen contain hundreds of millions or billions of parameters and achieve remarkable performance on language understanding and generation tasks.

Traditionally, adapting these models to downstream tasks requires full fine-tuning, where every model parameter is updated during training. Although effective, this approach is computationally expensive and memory intensive.

Recent research has introduced Parameter-Efficient Fine-Tuning (PEFT) methods that significantly reduce training requirements while preserving model performance.

1.2 Problem Statement

The objective of this project is to investigate whether LoRA and QLoRA can be used to efficiently adapt transformer-based models for sentiment classification while minimizing computational and memory costs.

2. Project Objectives

- Implement LoRA-based fine-tuning on BERT.
 - Implement QLoRA-based fine-tuning on Gemma 2B.
 - Analyze parameter efficiency.
 - Analyze memory efficiency.
 - Compare encoder-based and decoder-based adaptation strategies.
 - Evaluate model performance quantitatively and qualitatively.
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3. Dataset Description

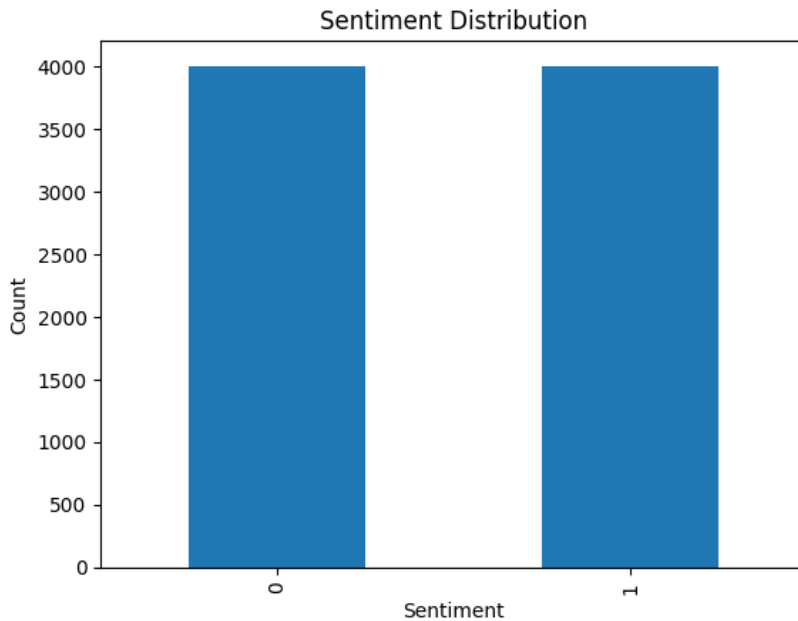
Dataset: IMDb Sentiment Dataset

Characteristics:

- Binary classification task
- Positive and Negative movie reviews
- Training Samples: 8,000
- Validation Samples: 2,000
- Test Samples: 10,000

The dataset provides a balanced sentiment distribution suitable for evaluating classification performance.

Figure 1: Dataset Distribution



4. Understanding PEFT

4.1 Full Fine-Tuning

In traditional fine-tuning, all model parameters are updated during training.

Advantages:

- High flexibility
- Strong task adaptation

Limitations:

- High GPU memory requirements
- Increased training cost
- Difficult deployment for large models

4.2 LoRA

Low-Rank Adaptation (LoRA) freezes the original model weights and trains only low-rank adapter matrices inserted into transformer layers.

Advantages:

- Reduced trainable parameters

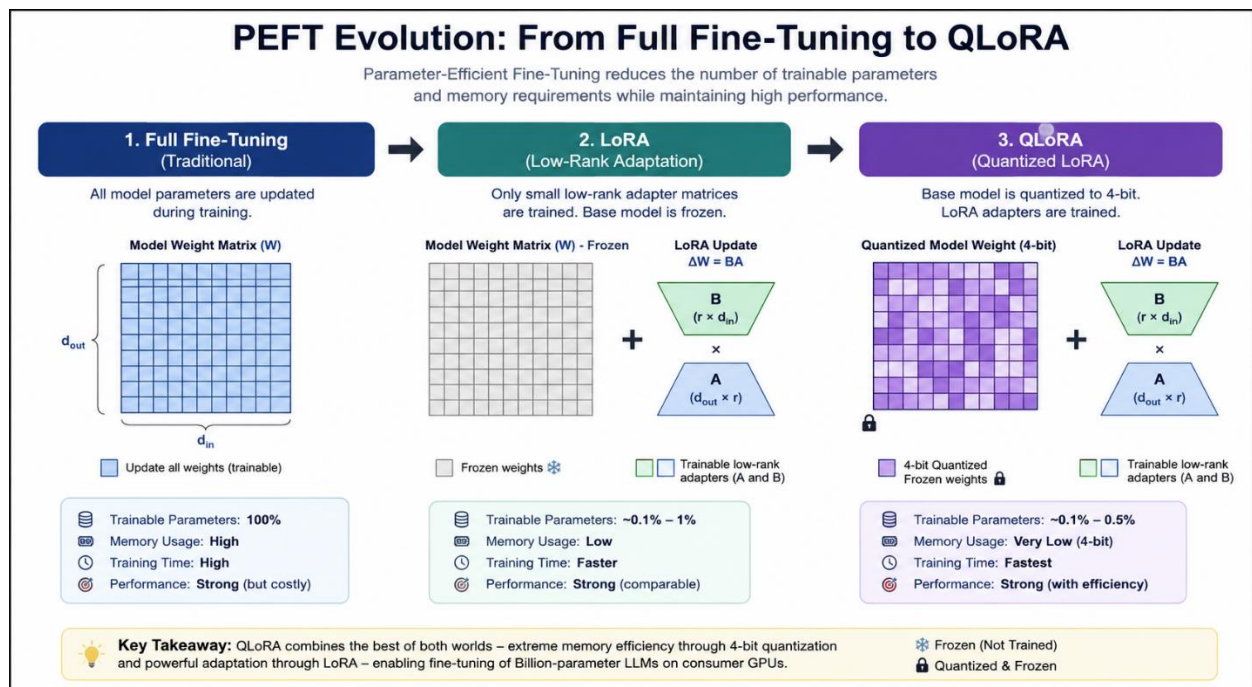
- Lower memory consumption
- Faster training

4.3 QLoRA

QLoRA extends LoRA by introducing:

- 4-bit quantization
- NF4 quantization format
- Quantized frozen backbone
- Trainable low-rank adapters

Figure 2: PEFT Evolution



5. BERT + LoRA Implementation

5.1 Model Selection

Model:

BERT Base Uncased

Characteristics:

- Encoder-only transformer

- 110 million parameters
- Pretrained on BooksCorpus and English Wikipedia

5.2 Data Preparation

The dataset was:

- Cleaned
- Tokenized
- Truncated to 512 tokens
- Converted into model inputs

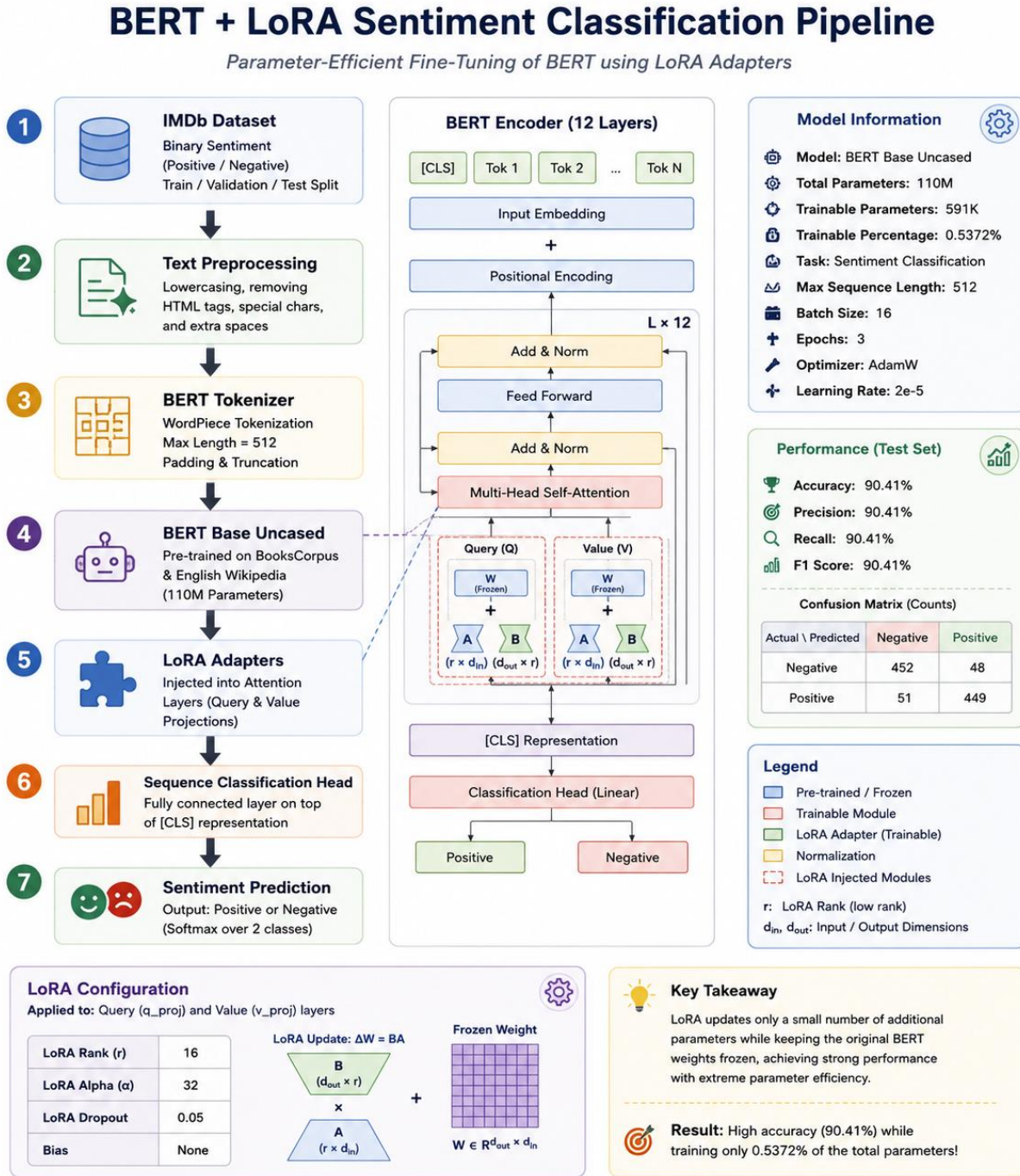
5.3 LoRA Configuration

Configuration:

- Rank (r): 16
- Alpha: 32
- Dropout: 0.05

LoRA adapters were applied to the attention layers.

Figure 3: BERT + LoRA Pipeline



6. Initial QLoRA Exploration on BERT

During the project, an attempt was made to apply QLoRA directly to BERT.

Although the model could be quantized and LoRA adapters were successfully injected, training encountered compatibility issues related to bitsandbytes 4-bit quantization and encoder-based architectures.

Rather than discarding the experiment, the debugging workflow was documented and retained within the repository.

This investigation provided valuable insight into the practical differences between encoder-based models and decoder-based Large Language Models.

7. Gemma 2B + QLoRA Implementation

7.1 Model Selection

Model:

Gemma 2B

Characteristics:

- Decoder-only transformer
- 2.62 billion parameters
- Instruction-tuning compatible

7.2 Instruction Formatting

The sentiment classification task was reformulated into an instruction-following format.

Example:

Instruction: Classify the sentiment of the following movie review.

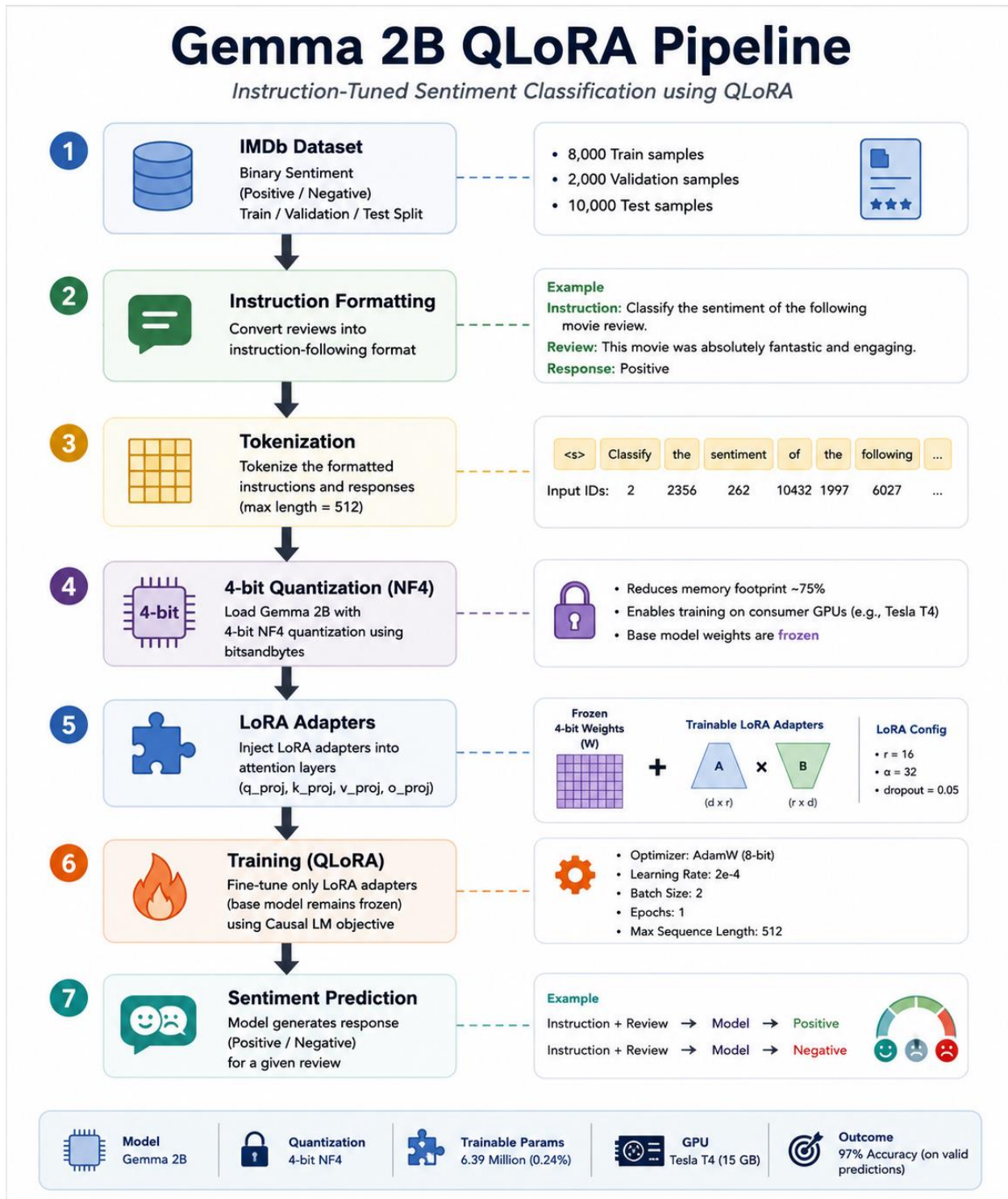
Review: This movie was excellent.

Response: Positive

7.3 QLoRA Configuration

- Quantization: 4-bit NF4
- LoRA Rank: 16
- LoRA Alpha: 32
- LoRA Dropout: 0.05

Figure 4: Gemma 2B QLoRA Pipeline



8. Training Configuration

BERT + LoRA

- Epochs: 3
- Max Length: 512
- Learning Rate: 2e-5

Gemma 2B + QLoRA

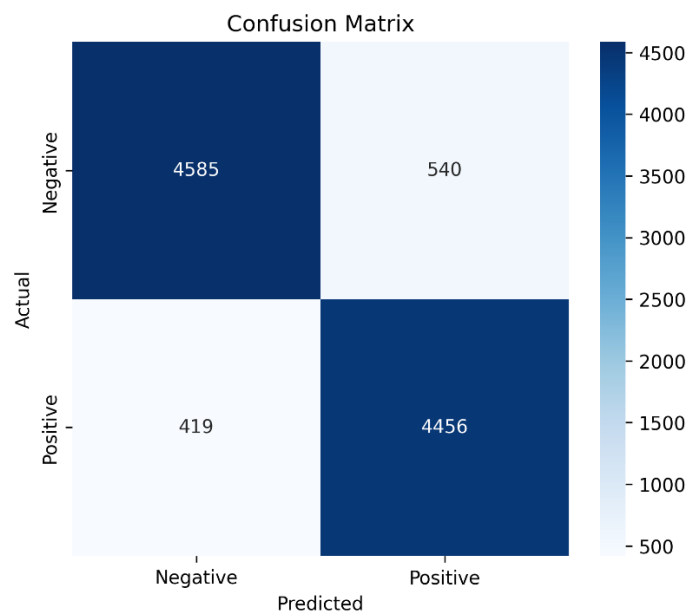
- Epochs: 1
 - Max Length: 512
 - Learning Rate: 2e-4
 - Quantization: 4-bit NF4
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9. Results and Evaluation

9.1 BERT + LoRA Results

Metric	Value
Accuracy	90.41%
F1 Score	90.41%

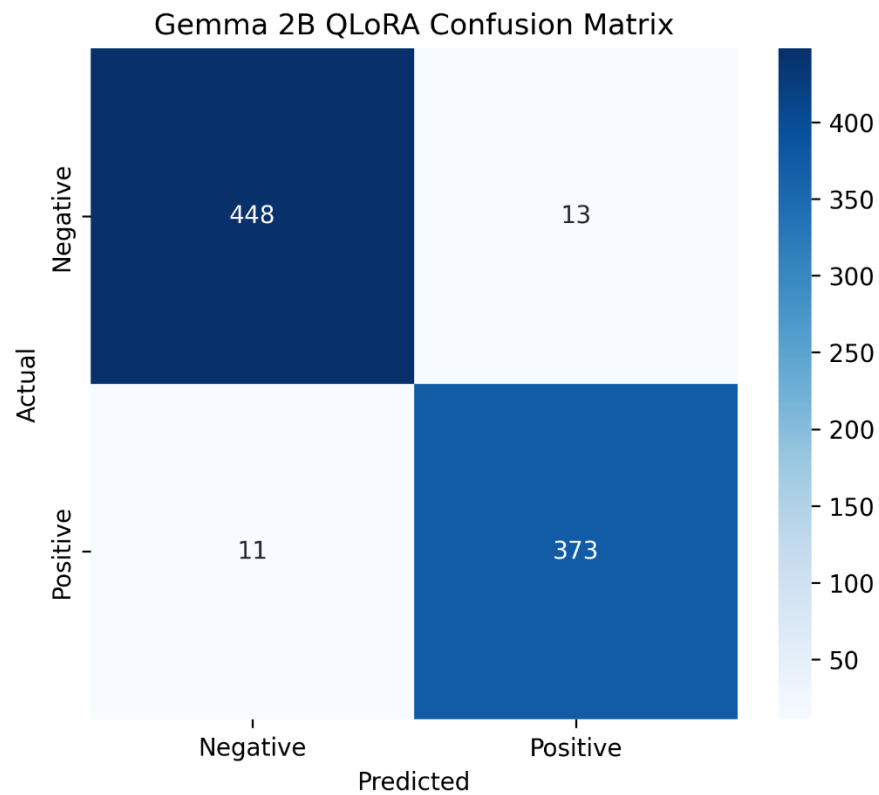
Figure 5: Confusion Matrix



9.2 Gemma 2B + QLoRA Results

Metric	Value
Accuracy	97.04%
Precision	97%
Recall	97%
F1 Score	97%

Figure 6: Confusion Matrix



10. Parameter Efficiency Analysis

Metric	BERT + LoRA	Gemma 2B + QLoRA
Total Parameters	110M	2.62B
Trainable Parameters	591K	6.39M
Trainable Percentage	0.5372%	0.2438%

Key Observation:

More than 99.75% of Gemma's parameters remained frozen during training.

11. Memory Efficiency Analysis

Gemma 2B was successfully trained using:

- Tesla T4 GPU
- 4-bit NF4 Quantization
- Approximately 3.3 GB memory allocation

This demonstrates the effectiveness of QLoRA for resource-constrained environments.

12. Key Learnings

- Understanding PEFT methodologies
 - Applying LoRA to encoder models
 - Applying QLoRA to decoder LLMs
 - Working with 4-bit quantization
 - Instruction tuning techniques
 - Memory-efficient adaptation of billion-parameter models
 - Practical debugging of PEFT workflows
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13. Future Work

Future extensions include:

- Domain adaptation using Mechanical Engineering datasets
 - Fine-tuning for educational tutoring tasks
 - Integration with the R.B.A.T. framework
 - Retrieval-Augmented Generation (RAG)
 - Agentic AI workflows
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14. Conclusion

This project successfully demonstrated the practical application of Parameter-Efficient Fine-Tuning techniques through LoRA and QLoRA.

The experiments showed that large language models can be effectively adapted using a very small fraction of trainable parameters while significantly reducing memory requirements.

The successful fine-tuning of Gemma 2B on a Tesla T4 GPU highlights the potential of QLoRA as a scalable and accessible approach for adapting modern LLMs in real-world environments.